

● MEDIUM

● ADVANCED MATH

# Equivalent expressions

7 questions · ~11 min

04

Q1

ADVANCED MATH

Which of the following is equivalent to the sum of  $3x^4 + 2x^3$  and  $4x^4 + 7x^3$ ?

- A.  $16x^{14}$
- B.  $7x^8 + 9x^6$
- C.  $12x^4 + 14x^3$
- D.  $7x^4 + 9x^3$

$$(x - 11y)(2x - 3y) - 12y(-2x + 3y)$$

Which of the following is equivalent to the expression above?

- A.  $x - 23y$
- B.  $2x^2 - xy - 3y^2$
- C.  $2x^2 + 24xy + 36y^2$
- D.  $2x^2 - 49xy + 69y^2$

Which of the following is an equivalent form of  $(1.5x - 2.4)^2 - (5.2x^2 - 6.4)$  ?

- A.  $-2.2x^2 + 1.6$
- B.  $-2.2x^2 + 11.2$
- C.  $-2.95x^2 - 7.2x + 12.16$
- D.  $-2.95x^2 - 7.2x + 0.64$

Which expression represents the product of  $x^{-6}y^3z^5$  and  $x^4z^5 + y^8z^{-7}$ ?

A.  $x^{-2}z^{10} + y^{11}z^{-2}$

B.  $x^{-2}z^{10} + x^{-6}z^{-2}$

C.  $x^{-2}y^3z^{10} + y^8z^{-7}$

D.  $x^{-2}y^3z^{10} + x^{-6}y^{11}z^{-2}$

$$4a^2 + 20ab + 25b^2$$

Which of the following is a factor of the polynomial above?

- A.  $a + b$
- B.  $2a + 5b$
- C.  $4a + 5b$
- D.  $4a + 25b$

Q6

GRID-IN

ADVANCED MATH

The sum of  $-2x^2 + x + 31$  and  $3x^2 + 7x - 8$  can be written in the form  $ax^2 + bx + c$ , where  $a$ ,  $b$ , and  $c$  are constants. What is the value of  $a + b + c$  ?

STUDENT-PRODUCED RESPONSE

|   |   |   |   |
|---|---|---|---|
|   |   |   |   |
| . | / | . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$gx = \frac{3}{5}x + \frac{7}{6}$$

$$hx = 6x - 5$$

The functions  $g$  and  $h$  are defined by the equations shown. Which expression is equivalent to  $gx \cdot hx$ ?

A.  $\frac{18x^2}{5} - \frac{35}{6}$

B.  $\frac{18x^2}{5} + \frac{27x}{11} - \frac{35}{6}$

C.  $\frac{18x^2}{5} - 4x - \frac{35}{6}$

D.  $\frac{18x^2}{5} + 4x - \frac{35}{6}$